

Adjoint-Faithful FNO Project Tracker

MITGCM baroclinic double-gyre benchmark

Status date: 28 July 2026 Canonical plan: AF_FNO_Project_Plan.tex

Current position. S0–S2 and the shared trajectory dataset are complete, and A0, A, and B remain frozen controls. Model C’s training-only C1b, 320-epoch C1c, and width-64 capacity tests all completed but were scientifically rejected because temperature or SSH did not beat persistence. GPU audit job 287581 checked every evaluated epoch and the exact late-checkpoint objective gradients; it found an already substantial, SSH-dominated increment gradient and supported late optimization stabilization rather than another capacity increase. Bounded GPU job 287583 retained loss v1 and reduced the learning rate from 0.0005 to 0.0001 after epoch 240; this schedule passed every 96-record memorization and reload criterion at epoch 315. The separately versioned 0.0025 increment-weight loss v2 was rejected.

The predeclared validation-only search is now complete. GPU jobs 287604/288466/289379/289915 executed all four from-scratch $10 \rightarrow 5 \rightarrow 3 \rightarrow 2 \rightarrow 1$ rounds and selected (24, 16), width 64, four layers, and 10% padding. Neither the capacity nor wall leakage trigger fired, so no six-layer or 20%-padding branch was opened. Final-seed array 290382 completed all three runs with exact three-step reload, but their SSH validation ratios were 1.00014/1.00383/1.03818; U, V, and temperature beat persistence for every seed. The immutable decision is therefore `scientifically_rejected_three_seed_gate`, not an operational failure. Model C is not frozen and inference is not authorized. Validation was used only under the frozen search contract; inference, intermediate-wind, response, and adjoint data remain sealed. Post-search data-adequacy job 290415 subsequently authorized a non-destructive two-times S0–S2 expansion while localizing a separate low-wind S1 training failure.

That expansion is now complete. MITGCM array 290439 produced all three exact ten-year continuations normally in 15m10s–15m31s. Jobs 290443/290444 built and independently validated the 11 GB, $3 \times 7200 \times 46 \times 62 \times 62$ `trajectories_v2` store, including a bitwise-identical v1 prefix and raw-MDS extension spot checks. Training-only coverage job 290446 then passed its predeclared two-times slow-field target: temperature/SSH state effective multipliers are 2.142/2.039 and increment multipliers are 2.106/2.253. Fresh v2 validation and inference remain unread. Successor contract `model_c_successor_training_v1` freezes an exact data-only control, an isolated width-64 4C Channel-MLP ablation, and only then the width-128/256-lift/256-projection/4C candidate. Both tasks in GPU array 290448 completed normally but failed only S1 SSH at 1.103435 and 1.003573 times persistence, respectively. Job 290597 completed width 128 normally and passed every training gate, including S1 SSH at 0.762683. Corrected validation contract SHA-256 56bb6ec59799...0cf27b is frozen. Split-1 array 290673 completed both missing replicas normally; all three seeds pass every training gate at worst regime/group ratios 0.762683/0.732270/0.839507. Fresh-v2 validation job 290738 completed operationally but rejected C scientifically: surface speed passes, while SST/surface-pressure RMSE-AUC ratios to persistence are 2.313/2.198 after good day-10 skill. Frozen training-only rollout-diagnosis job 291102 completed with exit zero and reproduced the same drift for every seed. Mean training day-10/AUC/day-90 ratios are 0.809/2.021/4.221 for SST and 0.552/1.867/3.873 for surface pressure, classifying an objective/checkpoint-gate mismatch. A baseline-visible Figure-4 companion now shows that SST loses persistence and climatology at day 20, while surface P/ρ loses persistence at day 30 and climatology at day 70. Exact-replay job 291164 then completed in 1h19m14s and reproduced step 14,880 bit for bit with zero training-history difference. None of the six late checkpoints passes the frozen long-horizon gate; diagnostic best step 14,400 still reaches day-90 SST ratios 3.669/5.513 to persistence/climatology and surface-pressure ratios 3.800/2.247. The applied classification is `objective_correction_required`.

Pushforward job 291612 completed normally in 20m37s. Its final step 1,920 is best by every frozen

priority and reloads exactly. Worst primary AUC improves from 2.366 to 1.692, worst slow-field lead ratio from 5.513 to 3.994, and ten-day worst regime/group ratio from 0.852 to 0.610, but the complete gate remains failed. Day-90 SST ratios are 2.659/3.994 and surface- P/ρ ratios are 3.222/1.905 to persistence/climatology. Replication and validation are not authorized.

Duration job 291616 completed normally in 51m31s and reproduced step 1,920 bitwise before adding all 3,840 low-rate steps. None of eight checkpoints passes. Diagnostic-best step 5,760 improves worst primary AUC/slow-field lead ratios to 1.597/3.609, but day-90 SST remains 2.402/3.609 and surface- P/ρ 2.922/1.727 times persistence/climatology. The contract therefore rejects undertraining as a sufficient explanation and applies the predeclared terminal-only-structure diagnosis.

Corrected truncated-unroll contract SHA-256 0c5aabadbe51...92e3c changed only the correction gradient graph. Initial job 291769 stopped before training because its validator used the wrong duration-checkpoint step-field name. Version 2 checked the actual schema without changing the objective, and V100 job 291778 completed normally in 20m01s. All four checkpoints fail. Diagnostic-best truncated step 1,920 reloads nine calls bitwise, but worst primary AUC/slow-lead ratios 1.933/3.701 are worse than the duration source at 1.597/3.609. No replica, fresh-validation, inference, or Model-D job is submitted; the next action is no longer another unconstrained temporal variant.

Frozen bias/projection job 292064 completed normally in 7m09s. Static 9ϵ explains 33.43%/27.21% of pooled SST/SSH day-90 error energy, while basin mean plus five increment EOFs explain only 4.57%/1.73%. Bias plus constraints improves the worst slow-field AUC ratio from 1.986 to 1.603 (19.25%), just below the frozen 20% threshold; conservation alone slightly worsens it to 1.996. Apply the feedback-amplification branch, retain the exact linear constraints and a modest bias term, and prioritize forecast-age supervision.

Loss-v3 job 292293 completed normally in 27m46s. Selected diagnostic step 960 is reload-exact and keeps the projection residuals below 1.17×10^{-10} . All ten-day groups pass, but the long gate does not: worst primary AUC improves from 1.597 to 1.462 while worst slow-field lead changes from 3.609 to 3.634. SST RMSE-AUC is 1.119/1.462 and surface- P/ρ is 1.364/0.615 times persistence/climatology. At the mandatory day-10/day-30 adjoint horizons, however, all Bire-primary fields and streamfunction beat both persistence and climatology. The next action is to freeze a separate prospective 10–30-day adjoint-readiness gate with damped persistence and anomaly-scale drift before any new validation read. The two-state/two-output model remains a bounded parallel long-rollout arm; no automatic training variant is authorized. All later archives remain sealed.

1 Status legend

Status	Meaning
Complete	Finished and checked against the stated acceptance evidence.
Active	Workstream is open and may contain completed substeps.
Next	Immediate executable step on the critical path.
Pending	Planned, but intentionally not started.

2 Milestone dashboard

ID	Milestone	Status	Evidence / exit condition
M0	Reproducible MIT-GCM build and segmented runner	Complete	S0 build job 283675; four-rank executable and immutable per-segment provenance.
M1	S0 control trajectory	Complete	100-year spin-up plus 10-year daily production; jobs 283676–283686; final data inventory verified.
M2	Tutorial validation	Complete	Validation job 283958; seven of seven gates passed; four project-facing figures and machine-readable report.
M3	S1/S2 forcing regimes	Complete	Low- and high-wind adjustment/production jobs 283959–283962 complete; daily inventories verified.
R0	Git provenance baseline	Complete	Clean first commit pushed to the project remote; generated data, upstream clones, builds, environments, and scheduler logs excluded.
M4	Shared ML dataset	Complete	Jobs 284041/284042 built and independently validated the 5.3 GB S0–S2 Zarr store: $3 \times 3600 \times 46 \times 62 \times 62$, centered/masked state, split hashes, and training-only normalizers.
M4b	Version-2 trajectory coverage	Complete	Jobs 290439/290443/290444 completed all exact extensions, the 11 GB successor store, and independent quality validation. Training-only job 290446 passed the predeclared two-times temperature/SSH effective-coverage target; fresh validation and inference remain sealed.
M5	A0 adapted Bire baseline	Complete	Jobs 284043/284050/284063/284065 completed the overfit, development, frozen model, and outputs. A0 improves held U/V forecasts but persistence remains better for temperature and SSH; metrics and figures are saved.
M6	A–D model ladder	Active	A0/A/B are frozen. Loss-v3 job 292293 passes all ten-day groups and both baselines at days 10/30, but fails the strict 10–90-day gate. Freeze the prospective horizon-matched adjoint-readiness gate next; keep long-rollout stability and the two-state arm parallel. Inference remains sealed.
M7	I1/I2 wind interpolation	Pending	Two unseen intermediate-wind trajectories generated and opened for inference only.
M8	Forward-response campaign	Pending	224 ten-day perturbed restarts: 128 training, 32 validation, and 64 inference.
M9	Blind adjoint evaluation	Pending	Frozen artifacts/checksums recorded before 12 true-adjoint tests are generated or inspected.
M10	Control-update verification	Pending	Three forward checks demonstrate whether emulator-gradient controls improve each objective.

3 Completed simulation record

3.1 S0: control wind, $\tau_0 = 0.100 \text{ N m}^{-2}$

Item	Recorded result
Execution	Build job 283675; spin-up jobs 283676–283685; production job 283686.
Model duration	100-year spin-up plus 10-year production: 110 model-years, using a 360-day year.
Production inventory	3,600 daily <code>dynState</code> records and 3,600 daily <code>surfState</code> records.
Measured size	4.6 GB in the raw MITGCM archive.
Validation	Job 283958; all seven tutorial gates passed.
Surface heat flux	Mean -1.9589 W m^{-2} .
Potential temperature	Final representative means: 17.6918°C at the surface, 8.3589°C at 305 m, and 2.4327°C in the abyss.
Gyre transport	First production-year extrema: $+32.679$ and -30.497 Sv .
Sea-surface height	First production-year range: -0.987 to $+0.762 \text{ m}$.

The temperature structure, double-gyre transport, sea-surface-height pattern, and late-spin-up trends are consistent with the intended official tutorial behaviour. The deep ocean remains the slowest-adjusting component, as expected; this is recorded as a scientific caveat rather than a failed gate.

3.2 S1 and S2: off-control wind regimes

Run	τ_0	Jobs	Schedule	Daily output	Size
S1	0.075 N m^{-2}	283959 / 283960	5-y adjust + 10-y production	3,600 <code>dynState</code> + 3,600 <code>surfState</code>	2.7 GB
S2	0.125 N m^{-2}	283961 / 283962	5-y adjust + 10-y production	3,600 <code>dynState</code> + 3,600 <code>surfState</code>	2.7 GB

Both branches restart from the validated S0 year-100 state. Their five-year adjustment periods are kept separate from the ten-year daily production records used by the ML pipeline.

3.3 S0–S2 exact trajectory-v2 continuations

Run	Array task	Elapsed	Iteration join		Recorded result
S0	290439_0	15m31s	2,851,128	to	3,600 daily field pairs, final pickup, normal end
			2,851,200		
S1	290439_1	15m28s	2,980,728	to	3,600 daily field pairs, final pickup, normal end
			2,980,800		
S2	290439_2	15m10s	2,980,728	to	3,600 daily field pairs, final pickup, normal end
			2,980,800		

The exact joins are one model day (72 iterations) after each immutable v1 endpoint. Every continuation contains 3,600 `dynState` and 3,600 `surfState` metadata/data pairs with no missing daily iteration. Logs end in `STOP NORMAL END`; the floating-point status flags printed at shutdown match the previously accepted production runs and accompany no NaN, fatal, or model error. The three new pickup data SHA-256 values begin `c949...315fb`, `06be...8087`, and `e5a8...1744`.

3.4 Retired high-resolution Bire reconstruction

The earlier 0.25° Bire-oriented spin-up, formerly Slurm job **283514**, was cancelled by the project owner on 21 July 2026. Its scratch tree, compiled build, MITgcm input deck, scheduler wrappers, conversion/training wrappers, and generated reports were subsequently deleted. This campaign is closed and is not a prerequisite for A0 or AF-FNO. Only recovered source and documentation that provide architectural evidence for A0 are retained.

4 Data locations and provenance

Asset	Canonical location
S0 project entry point	<code>outputs/af_fno/S0/</code>
S0 result catalog	<code>outputs/af_fno/S0/results_catalog.json</code>
S0 validation	<code>outputs/af_fno/S0/tutorial_validation/</code>
S0 provenance copies	<code>outputs/af_fno/S0/provenance/</code>
S0 raw archive	<code>/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S0/</code>
S1 raw archive	<code>/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S1/</code>
S2 raw archive	<code>/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm/S2/</code>
Shared ML dataset	<code>/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v1.zarr</code>
Dataset manifest	<code>/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v1.manifest.json</code>
Dataset quality report	<code>/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v1.quality.json</code>
Trajectory-v2 raw continuations	<code>/bigscratch/mjalabert314/bire_james25_repro/af_fno/mitgcm_v2/</code>
Trajectory-v2 dataset	<code>/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v2.zarr</code> ; metadata SHA-256 <code>ae7be47c4569...78b8cc</code>

Asset	Canonical location
Trajectory-v2 quality gate	/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v2.quality.json; SHA-256 a6f7e5234ace...cf752
Trajectory-v2 effective coverage	/bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v2_coverage_v1/; report SHA-256 7758cfb64e73...18e5
Pressure-postprocessor validation	outputs/af_fno/pressure_validation_v1/pressure_validation.json
Streamfunction consistency audit	V1 collocation-warning record under outputs/af_fno/streamfunction_consistency_v1/; corrected collocated v2 contract config/streamfunction_consistency_v2.json, SHA-256 88ff2f20fe1a...dea170; report/arrays SHA-256 14fa2bfb2d2b...eae9a7/f807d3025186...c55773; figures and manifest-content SHA-256 d62eed150a13...e6162 under outputs/af_fno/streamfunction_consistency_v2/
A0 overfit artifacts	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A0/overfit_v1/
A0 development artifacts	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A0/development_v1/
A0 frozen artifacts	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A0/final_v1/
A0 figures / numerical outputs	outputs/af_fno/A0/final_v1/
A0 paper-style output package (job 284150 complete)	outputs/af_fno/A0/paper_style_v1/
Model A overfit artifacts	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A/overfit_v1/
Model A development artifacts (job 284152)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A/development_v1/
Model A frozen artifacts (job 284153)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/A/final_v1/
Model A figures / numerical outputs (job 284155 complete)	outputs/af_fno/A/final_v1/
Model B overfit artifacts (job 284302 complete)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/B/overfit_v1/
Model B development profiles (jobs 284304–284306 complete)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/B/development_*_v1/
Model B frozen artifacts (job 284342 complete)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/B/final_v1/
Model B figures / numerical outputs	outputs/af_fno/B/final_v1/
Model C training-only diagnostics (jobs 284850/284857)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/training_diagnostics_v1/ and /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/training_diagnostics_v2/

Asset	Canonical location
Model C GPU loss-gradient calibration (job 284858)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/loss_calibration_v1/
Model C CPU loss-gradient replica (jobs 284860/284864/285116)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/loss_calibration_cpu_v1/
Model C frozen loss manifest	config/model_c_loss_v1.json
Model C memorization/reload gate (job 285192)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_v1/
Model C controlled duration gate (job 285265)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_duration_v2/
Model C width-64 GPU capacity gate (job 285325)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_width64_gpu_v1/
Model C late-checkpoint audits (job 287581)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/late_checkpoint_objective_audit_width32_v1/ and /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/late_checkpoint_objective_audit_width64_v1/
Model C accepted optimization manifest	config/model_c_optimization_v1.json
Model C accepted memorization gate (job 287583 task 0)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_v1_lr_decay_v1/
Model C rejected loss-v2 manifest / gate (job 287583 task 1)	config/model_c_loss_v2.json and /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/overfit_increment_loss_v2/
Model C CPU fallback scaling (job 287556)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/benchmarks/C/cpu_scaling_width64_v2/
Model C validation-search contract	config/model_c_validation_search_v1.json; SHA-256 9e1d44299ae6...c4c6f6
Model C validation-search stages	Jobs 287604, 288466, 289379, and 289915; immutable root /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/validation_search_v1/; stage-manifest SHA-256 values 40b54530d7d9...12d2f05, a2d639f8940c...d5e382, 9bbd4fc4ce3c...f30e17, and 453c05ec408f...2fea2
Model C rejected three-seed gate (job 290382)	/bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/validation_search_v1/final_seed_gate/; decision SHA-256 e19d502442fc...72155, freeze-contract SHA-256 a44622cee685...73628
Model C data-adequacy contract / job 290415	config/model_c_data_adequacy_v1.json; SHA-256 e185301947b4...53c84a9; report /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/data_adequacy_v1/model_c_data_adequacy_report.json, SHA-256 be672face8c4...4e307ee

Asset	Canonical location	
Trajectory-v2 expansion contract	config/trajectories_v2_expansion.json, 7ee2ae10330a...fe045ed; three exact ten-year extensions under the new mitgcm_v2/ scratch root	SHA-256
Trajectory-v2 coverage contract / job 290446	config/trajectories_v2_coverage_audit.json, e539ca1bb61a...9197f36; immutable report root /bigscratch/mjalabert314/bire_james25_repro/af_fno/datasets/trajectories_v2_coverage_v1/	SHA-256
Model C successor-training contract / jobs 290448/290597	config/model_c_successor_training_v1.json, 3e0a300e73ec...159285; immutable checkpoints/reports under /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/successor_training_v1/; lightweight results under outputs/af_fno/C/successor_training_v1/	SHA-256
Model C successor validation / jobs 290673/290738	Corrected config/model_c_successor_validation_v2.json, 256 56bb6ec59799...0cf27b; complete rejected report SHA-256 9853ea7aff21...663fb; immutable scratch root /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/successor_validation_v1/; lightweight training and validation evidence under outputs/af_fno/C/successor_validation_v1/	SHA-256
Model C validation figures	Five hash-bound PNGs and figure_manifest.json under outputs/af_fno/C/successor_validation_v1/fresh_validation/; manifest content SHA-256 8b24b6af0686...a0845f	
Model C rollout diagnosis / job 291102	config/model_c_rollout_diagnosis_v1.json, d0ecee0db2b2...ff7ba; complete report/arrays SHA-256 7efdb0726125...171374/c3c403ab81b2...5f255; immutable scratch root /bigscratch/mjalabert314/bire_james25_repro/af_fno/models/C/rollout_diagnosis_v1/; three figures and lightweight summary under outputs/af_fno/C/rollout_diagnosis_v1/	SHA-256
Model C exact-replay checkpoint audit / job 291164	Completed in 1h19m14s; exact step-14,880 replay, no passing checkpoint, diagnostic best step 14,400. Contract SHA-256 494ce9de6688...06682; report/arrays SHA-256 7fd670f482c5...940367/7983cb5e0c74...ee9227; published root outputs/af_fno/C/checkpoint_replay_audit_v1/	SHA-256
Model C pushforward objective / job 291612	Completed scientific rejection in 20m37s; exact selected step 1,920, worst primary AUC/slow-lead ratios 1.692/3.994. Contract SHA-256 406ee5248fb4...bccc2; report/arrays SHA-256 631cec8e5f49...ebd07/a4469e3c2d3d...b0855; published root outputs/af_fno/C/pushforward_objective_v1/	SHA-256
Model C pushforward duration / job 291616	Completed in 51m31s with bitwise pushforward-v1 replay and exact reload; all eight checkpoints fail, and diagnostic-best step 5,760 has worst primary AUC/slow-lead ratios 1.597/3.609. Contract SHA-256 4867fa77d68e...78c79; report/arrays SHA-256 86bf00b800d6...e9d7/46129f2653b7...7bfc; published root outputs/af_fno/C/pushforward_duration_v1/	SHA-256

Asset	Canonical location
Model C truncated-unroll objective / jobs 291769/291778	Job 291769 stopped before training on a checkpoint-field schema mismatch. Corrected job 291778 completed in 20m01s but failed at diagnostic-best worst primary AUC/slow-lead ratios 1.933/3.701. Contract SHA-256 0c5aabadbe51...92e3c; report/arrays SHA-256 86f7981e7a0a...b5b830/0c5222cd6938...d716; figure/CSV/summary under outputs/af_fno/C/truncated_unroll_objective_v2/, manifest-content SHA-256 ccccd8e757495...ae4c03
Model C slow-field bias/projection audit / job 292064	Completed in 7m09s; feedback-amplification classification, fixed-bias SST/SSH explained energy 33.43%/27.21%, and 19.25% combined worst-AUC reduction. Contract SHA-256 fd7d39f5c6a4...05428; report/arrays SHA-256 e66b263c826c...0f90/4e7f73c2384e...8547; plots/CSV/summary under outputs/af_fno/C/slow_field_bias_projection_v1/, manifest SHA-256 0a2b4fbc8411...c4d
Model C rollout-conditioned loss v3 / job 292293	Completed in 27m46s; diagnostic step 960, all ten-day groups pass, strict long gate fails at worst primary AUC/slow-lead ratios 1.462/3.634. Contract SHA-256 37f1157851ac...5a08; report/arrays/checkpoint SHA-256 e9af34326c36...7ba11/522fe6145735...edff/e9f962878cb5...970b; two plots, CSV, summary, and manifest c323a18285f1...0a6d under outputs/af_fno/C/rollout_conditioned_loss_v3/
Model C Bire-style figures / job 291134	Completed frozen descriptive contract config/model_c_bire_figures_v1.json, SHA-256 791c84f12c2d...eb696b; report/arrays SHA-256 b704b921444d...e300ac5/a73ac9bf2cc9...20b1; two PNGs and hash manifest under outputs/af_fno/C/bire_figures_v1/, manifest-content SHA-256 258be6fabe13...827f
Model C Figure-4 baseline companion	Frozen config/model_c_bire_baseline_zoom_v1.json, SHA-256 2a2324090b83...b2a4a; full/zoom SST and surface- P/ρ figure plus complete CSV under outputs/af_fno/C/bire_baseline_zoom_v1/, manifest-content SHA-256 01433936e406...99063
Model C per-lead streamfunction maps / job 291135	Completed descriptive contract config/model_c_bire_streamfunction_leads_v1.json, SHA-256 ccb965e58bdc...ae1934c; report/arrays SHA-256 71c6e1dc1f9a...89c206/b4126494d9c0...3bcd48; eight PNGs and manifest-content SHA-256 3b74d759d551...450ac7 under outputs/af_fno/C/bire_streamfunction_leads_v1/

The S0 project directory contains a symbolic link named `raw_mitgcm`; bulk output is not duplicated in `/home`. The catalog records counts, final pickup, hashes, executable provenance, and canonical paths. The shared-dataset manifest seals the three source archives, the fixed split, and normalizer hashes; its independent quality report records finite/mask/spectral checks, one-day increment ranges, and a fresh normalization recomputation.

5 Frozen project decisions

- The active scientific benchmark is the official 1° , $62 \times 62 \times 15$ baroclinic double gyre. Exact recovery of the paper's hidden 0.25° numbers is outside the primary claim; qualitative trends are the reproduction target.

- Model calendars use 360 days per year. The time step is 1,200 s, or 72 steps per model day.
- Standard trajectory production uses four MPI ranks. Long runs are split into restart-safe jobs that fit the two-hour `short` partition.
- S0–S2 share physics, grid, preprocessing, state definition, splits, and metrics. Only the prescribed forcing regime differs.
- PHIHYD remains an exact configured hydrostatic reconstruction from all 15 temperature levels and SSH. Streamfunction is labeled the U-derived barotropic transport streamfunction. The collocated raw-face U/V audit passes daily and annual path consistency in S0–S2 by orders of magnitude, but the linear-free-surface instantaneous-divergence caveat remains explicit.
- True MITGCM adjoints are sealed test data. They are excluded from training, early stopping, architecture selection, hyperparameter tuning, perturbation-scale selection, and loss selection.
- All models must use the same immutable train/validation/test partition and training-only normalization statistics so architecture and loss comparisons remain interpretable.
- Model C validation-search v1 is closed as a scientific rejection. Its three-seed decision records `configuration_frozen=false` and `inference_authorized=false`; no later-data archive may be opened on the strength of the near-threshold SSH results.
- Git stores source, configuration, tests, small manifests, and project-facing documentation. Raw/reduced arrays, checkpoints, upstream clones, virtual environments, compiled trees, and scheduler logs remain external to Git.

6 Baseline and forward-selection ladder

Decision added 21 July and revised 23 July 2026. Establish an adapted Bire paper-architecture baseline, A0, before developing the modern NeuralOperator A–D sequence. This tests whether the published architectural idea recovers the paper’s qualitative wind and forecast trends on the available tutorial configuration. Keep B frozen, select C only for forward fidelity, and add local-response supervision only in D.

A0 is a controlled adaptation, not a claim of exact historical reproduction. It keeps the recognizable Bire choices—channel-diagonal/separable spectral convolution, explicit pointwise channel mixing, three Fourier blocks, hidden width 128, positional encoding, and one-step state loss. Resolution-equivalent Fourier truncation uses 16 modes on the 62×62 grid. The frozen A0 did not use the paper code’s stride-8 coarse pretraining; this is a recorded adaptation, not an inferred reproduction.

Changes forced by this benchmark must be disclosed: 46 Markov-state channels (centered U, V, temperature, and SSH), the 62×62 grid, ten production years per regime, wind stress as an input, and a ten-day primary comparison horizon. The obsolete public repository will be used as architectural evidence, not executed unchanged. Pressure is not a forgotten channel: evaluation contract v2 reconstructs MITGCM PHIHYD at levels 0, 7, and 14 from predicted temperature and SSH using the configured linear EOS and exact finite-difference hydrostatic operator. Its archived S0 validation passes with $3.89 \times 10^{-4} \text{ m}^2 \text{ s}^{-2}$ maximum error. The velocity-derived circulation diagnostic is independently verified under `config/streamfunction_consistency_v2.json`: daily U/V path p90 relative RMSE is at most 3.25×10^{-5} , annual absolute mismatch is $4.74\text{--}6.00 \times 10^{-5} \text{ Sv}$, correlation is effectively one, and all boundary normal transports are zero. This supports the U-derived transport streamfunction strongly while retaining the free-surface caveat.

The comparison ladder is therefore:

Persistence / climatology $\rightarrow$ A0: adapted Bire $\rightarrow$ A: modern state-only
 $\rightarrow$ **B: forward losses $\rightarrow$ C: forward-selected $\rightarrow$ D: AF-FNO response supervision.**

Every transition changes a declared element, making gains attributable rather than confounding the dataset, architecture, and derivative-aware objective at once.

7 Immediate execution sequence

Current next step: freeze the 10–30-day adjoint-readiness gate. Completed loss-v3 job 292293 selected reload-exact step 960. Its strict 10–90-day gate fails, so it cannot support a long-emulator claim, but every Bire-primary field plus streamfunction beats persistence and climatology at both mandatory adjoint horizons. Before reading any new validation metric, freeze the exact day-10/day-30 leads, damped-persistence baseline, anomaly-scale drift bound, three-seed or bootstrap rule, and linear constraint tolerances. Only a prospectively passing short gate may open response and adjoint work. Keep the 90-day rollout-stability track and the bounded two-state/two-output arm separate; do not launch another loss-weight, duration, or capacity variant.

The starting C architecture is B’s dense four-layer, width-32 residual FNO, but its objective gives equal status to U, V, temperature, and SSH; adds training-only per-channel increment scaling; and compares tapered spectra of ten-day increments on the exact wet rectangle. The bounded later search contains modes (12, 12), (16, 16), (16, 24), (24, 16), (24, 24), widths 32/64, and conditional padding/layer tests. Width 32 has 630,262 parameters; width 64 has 2,443,542 and is not justified unless capacity diagnostics require it. C0 jobs 284850/284857 found a 147-fold range in per-channel increment RMS, only 278–398 effective increment samples across regimes, and substantially more velocity-increment energy in meridional than zonal added modes. Complex-safe job 285116 froze loss-v1 increment/rollout/spectral/boundary coefficients at 0.001/0.15/10^{−5}/0.065. C1b, C1c, and width-64 training-only results showed that more epochs or parameters did not remove the SSH failure. Late-checkpoint audit 287581 found weighted increment-gradient norms 0.422/0.406 times state at widths 32/64 and SSH-state gradient cosines 0.966/0.927. In bounded job 287583, raising the increment weight to 0.0025 failed, while preserving loss v1 and reducing the learning rate fivefold after epoch 240 passed every memorization criterion at epoch 315. The accepted schedule, report, checkpoint, and hashes are sealed in `config/model_c_optimization_v1.json`. Search contract v1 then ran exactly as frozen: four from-scratch rounds at 25/50/75/100% chronology and 1,920/3,840/5,760/7,680 steps. The (24, 16), width-64 finalist improved its worst validation group from 2.168 to 1.483, 1.192, and 1.00014 as chronology increased, but never crossed the strict gate. The selected architecture’s full-resource U/V/temperature/SSH ratios were 0.1473/0.3547/0.6597/1.00014. Neither conditional architecture trigger fired. Fresh final seeds 20260723/20260724/20260725 all reloaded exactly and all missed only SSH, at 1.00014/1.00383/1.03818. The v1 search is exhausted and scientifically rejected; it does not authorize inference, D, more architecture/loss tuning on the same validation evidence. Separately frozen data-adequacy contract SHA-256 `e185301947b4...53c84a9` was then executed in GPU job 290415. Its immutable report SHA-256 `be672face8c4...4e307ee` authorizes only a versioned, two-times coverage expansion; it does not reverse the v1 rejection or open inference.

1. Preserve the completed stage manifests, candidate reports/checkpoints, final-seed artifacts, and rejection decision without mutation. Treat all GPU tasks as operational successes and the gate outcome as a scientific rejection.

2. Preserve the completed data-adequacy report. Its four checks pass: aggregate training fit, strictly improving chronology, positive SSH validation–training gaps in all three regimes, and fewer than 20 effective slow-state samples. The S1 training failure prevents interpreting this as a data-only diagnosis.
3. Preserve completed jobs 290439/290443/290444/290446 and their immutable raw, dataset-quality, and effective-coverage evidence. The slow-field two-times target passes; do not use the less stable velocity integral-time ratios as a tuning target.
4. Preserve completed training-only successor jobs 290448/290597 under contract SHA-256 `3e0a300e73ec...159285`. The unchanged width-64 data control and isolated 4C channel-mixing test both failed only S1 SSH; 4C reduced the full-training ratio from 1.103435 to 1.003573 and improved every aggregate group. Width 128 passes every regime/group at worst ratio 0.762683 and every reload/stability check.
5. Preserve completed jobs 290673/290738 and the exact fresh-validation rejection. Surface speed passes every baseline; SST and surface pressure have good day-10 skill but fail persistence over the 10–90-day RMSE curve. Do not reinterpret the amplitude-only 180-day training check as trajectory skill.
6. Preserve completed training-only rollout-diagnosis job 291102 under contract SHA-256 `d0ecee0db2b2...ff7ba`. Every seed has good day-10 slow-field skill but failed 10–90-day AUC and day-90 skill on training chronology, so apply the predeclared objective/gate-mismatch classification. Do not generate more data or open inference.
7. Preserve exact-replay checkpoint-audit job 291164 under contract SHA-256 `494ce9de6688...06682`. Step 14,880 and its six-row history reproduce exactly, but no original checkpoint passes; step 14,400 is diagnostic best. Apply `objective_correction_required`, not a checkpoint-only correction.
8. Preserve rejected pushforward-objective job 291612 under contract SHA-256 `406ee5248fb4...bccc2`. Step 1,920 improves every displayed curve and all source priorities but still fails both slow-field baselines at long lead. Do not replicate or validate it.
9. Preserve completed duration job 291616 under contract SHA-256 `4867fa77d68e...78c79`. Require bitwise replay of pushforward step 1,920 passed, but none of eight extension checkpoints passes. Retain diagnostic-best step 5,760 and the published SST/ P/ρ curves; classify terminal-only supervision as structurally insufficient and do not validate it.
10. Preserve completed corrected truncated-unroll contract SHA-256 `0c5aabadbe51...92e3c`. Starting from duration step 5,760, job 291778 supervised three consecutive differentiable calls over the frozen early/late slow-field windows and reloaded nine calls bitwise. It failed every checkpoint and worsened worst primary AUC/slow-lead ratios to 1.933/3.701. Stop before validation and reassess from the saved lead/training curves.
11. Keep Model C’s temporal interface explicitly classified as one-input/one-output: 51/46 are channel counts, while each call sees one state and predicts one ten-day residual. Samudra’s two-past-state/two-future-state design, which Bire proposed only as future work, is a bounded candidate rather than an immediate replacement. Compare it prospectively against a simpler long-rollout objective/checkpoint correction at matched data and controlled compute.
12. Preserve streamfunction-consistency contract v2 SHA-256 `88ff2f20fe1a...dea170`. It corrects the v1 C-grid face-collocation error without changing the fixed year or thresholds. All S0–S2 daily/annual U/V path, averaging, amplitude, and impermeable-boundary checks pass. Use the U-derived transport-streamfunction label and retain the instantaneous free-surface-divergence caveat.
13. Preserve Bire-style figure contract SHA-256 `791c84f12c2d...eb696b`. It fixes a descriptive 1° streamfunction comparison and 15-member, 0–200-day $\Delta t = 10$ RMSE curves on S2 fresh validation. Persistence repeats the initial condition; climatology is a pointwise split-1 training mean. The figures may characterize the rejected model but may not tune or unseal inference. Job 291134 completed:

streamfunction is excellent through day 40, whereas SST and surface pressure leave their requested axes at days 100 and 160 as finite long-rollout drift grows. Preserve both the fixed-axis plots and off-scale numerical archive. Preserve the separate `outputs/af_fno/C/bire_baseline_zoom_v1/` addendum: it changes no source result and makes the SST/ P/ρ baseline crossings visible, with a complete CSV for every plotted lead.

14. Preserve per-lead streamfunction-map contract SHA-256 `ccb965e58bdc...ae1934c`. It fixes eight separate maps at days 20–90 for the existing S2 start 6335. State panels share one scale, while each difference panel has a disclosed lead-specific scale plus numerical RMSE/max error. It is descriptive only; do not use the maps to select a checkpoint or unseal inference. Job 291135 completed with all predictions finite: relative streamfunction RMSE rises only from 0.700% at day 20 to 2.211% at day 90, showing that barotropic circulation remains much better preserved than SST and surface pressure.
15. Keep inference, I1/I2, response, and adjoint data sealed. Open them only if an explicitly versioned Model C successor passes a prospectively declared forward gate; train D only from such an accepted design.

8 Remaining simulation and evaluation programme

Work package	Count	Role
S0–S2 trajectory-v2 extensions	3 complete	Exact continuations, conversion, independent quality control, and effective-coverage audit completed in jobs 290439/290443/290444/290446.
I1/I2 trajectories	2	Intermediate wind amplitudes; inference-only interpolation test.
Perturbed forward responses	224	Ten-day local responses: 128 training, 32 validation, 64 inference.
Mandatory true adjoints	12	Three smooth objectives $\times$ two horizons (10/30 days) $\times$ two unseen states.
Forward control checks	3	Test one emulator-gradient control update for each objective.
Optional true adjoints	6	Ninety-day extension only if mandatory criteria are met.

The expanded programme contains 247 scientific integrations in total: the original five trajectory runs, three authorized S0–S2 extensions, 224 perturbed forward responses, 12 mandatory adjoints, and three forward control checks. The extensions add 30 model-years, taking trajectory accounting from 142 to 172 model-years.

The mandatory adjoints alone are the ground-truth gradient dataset. They evaluate: an interior Gaussian SSH average, a smooth western-boundary SSH average, and an area-weighted barotropic gyre-strength objective. The emulator, evaluation code, pass criteria, and cryptographic hashes must be frozen before these results are opened.

9 Risks, corrections, and controls

Risk / issue		Control or recorded correction
Scratch is not archival		Raw S0–S2 data live on <code>/bigscratch</code> . Copy them to durable project storage before the cluster retention deadline; retain catalogs, hashes, manifests, and figures under the project tree.
Trajectory-year arithmetic		The original durations sum to 142 , not 132, model-years: $110 + 15 + 15 + 1 + 1 = 142$. The authorized three ten-year extensions raise the active compute/storage accounting to 172 model-years.
Historical-code incompleteness		Label A0 “adapted paper architecture” and record all forced changes; do not claim a bitwise or numerical reproduction of Bire et al.
Data leakage		Freeze chronological indices and derive normalization only from training years. Keep I1/I2 inference-only.
Adjoint leakage		Enforce the sealed-adjoint contract and checksum frozen model and evaluation artifacts before generating or viewing adjoint results.
Deep-ocean drift		Report depth-dependent trend diagnostics with forecast results; avoid implying full abyssal equilibrium from upper-ocean stationarity.
Large job count		Pack short response integrations into auditable Slurm arrays or bundles while preserving one manifest/result record per scientific integration.
Cross-HPC portability		UCSB absolute paths, modules, accounts, and partitions are not NASA-portable. Introduce site profiles, rebuild dependencies and executables on the destination system, transfer data separately, and pass smoke/restart tests before production.

10 How to update this tracker

Update this document after a milestone changes state, a scientific decision changes, or a material dataset/model artifact is created. For each update:

1. advance the status date and change only milestones supported by durable evidence;
2. record Slurm job IDs, model-time duration, output counts, storage location, and validation result for every completed simulation;
3. add artifact paths and checksums where available; never treat a scratch pathname alone as an archival record;
4. record decision changes in both the relevant section and the change log;
5. regenerate `Project_tracking.pdf` and check the LaTeX log for undefined references, overfull boxes, or missing files.

Operational details and evolving status belong here. Scientific definitions, acceptance criteria, and the intended experimental design belong in `AF_FNO_Project_Plan.tex`; keep the two documents consistent without duplicating extensive derivations.

11 Change log

Date	Type	Change
2026-07-21	Tracker created	Consolidated S0–S2 execution evidence, validation metrics, data locations, active decisions, remaining simulations, and risks.
2026-07-21	Method decision	Added A0, an adapted Bire paper-architecture baseline, before models A–C.
2026-07-21	Accounting correction	Corrected total trajectory duration from 132 to 142 model-years.
2026-07-21	Infrastructure	Prepared the source tree for Git: removed disposable Slurm/LaTeX/Python artifacts, strengthened ignore rules, replaced the paper-only README with the AF-FNO project overview, and documented NASA portability. The first remote commit remains R0.
2026-07-21	Campaign retired	Recorded cancellation of job 283514 and removed the obsolete 0.25° MITgcm scratch data, build, input deck, job wrappers, and reports. Retained only Bire-derived code evidence required for A0.
2026-07-21	Dataset conversion	Added and locally tested the S0–S2 MDS-to-Zarr converter, including declared C-grid centering, wet-mask handling, chronological split codes, and training-only normalizers. All 39 available tests passed (one optional ML test skipped); Slurm job 284041 was submitted to create and validate <code>trajectories_v1.zarr</code> .
2026-07-21	M4 complete	Conversion job 284041 completed in 1m32s and produced the 5.3 GB $3 \times 3600 \times 46 \times 62 \times 62$ store. Independent job 284042 completed in 48s; it verified finite values, masks, split contract, spectrum, daily continuity, and recomputed the training-only normalizers to 3.12×10^{-7} mean and 1.56×10^{-7} scale maximum absolute error.
2026-07-21	A0 gate started	Added and smoke-checked the adapted A0 loader/trainer: 46 normalized state inputs, one normalized wind input, 46 normalized next-state targets, the recovered three-block separable FNO with 16 modes, masked paper-style MSE+MAE loss, and bitwise save/reload acceptance. Submitted the 96-sample GPU overfit gate as job 284043.
2026-07-21	A0 overfit passed	Job 284043 completed in 6m11s on one V100 GPU. Its 96 regime-balanced, training-only ten-day pairs reduced the masked normalized loss from 1.04999 to 0.002128 at learning rate 0.01; no fallback was needed, and saved/reloaded inference was bitwise identical. Checkpoint SHA-256: 30406885a1d4...295c83b0e9.

Date	Type	Change
2026-07-21	A0 development started	Added the sealed-split development runner with chunk-aware chronological batches and persistence comparison. Submitted V100 job 284050 for 12 epochs over 7,530 training pairs and 780 validation pairs; this is a development gate, not the frozen final A0 model.
2026-07-21	A0 schedule frozen	Development job 284050 completed in 18m12s. Its best validation loss was 0.001841 versus persistence 0.014453, with bitwise checkpoint reload; the minimum occurred at epoch 10. Froze the final configuration (10 epochs, batch 8, learning rate 0.01, unchanged paper-style optimizer/loss/seed) and submitted its single final realization as V100 job 284063.
2026-07-21	A0 model frozen	Final job 284063 completed in 14m46s. It preserved bitwise checkpoint reload and obtained losses 0.001841/0.002272 on held validation/inference versus persistence 0.014453/0.014357. Checkpoint SHA-256: b879b64b8cc7...3663be18f. Submitted job 284065 to save the numerical and figure outputs in the project-facing output tree.
2026-07-21	A0 baseline closed	Evaluation job 284065 completed in 1m09s and saved <code>a0_metrics.json</code> , <code>a0_metrics_arrays.npz</code> , and three PNG figures under <code>outputs/af_fno/A0/final_v1/</code> . Held one-step A0/persistence RMSE ratios are 0.395 (U), 0.912 (V), 4.771 (temperature), and 13.612 (SSH); 10–100-day rollout figures show the same mixed state-variable result. M5 is complete without a claim that A0 beats persistence for every component.
2026-07-21	A0 paper-style evaluation complete	V100 job 284150 completed in 53s without retraining or changing A0 and saved a distinct output package. Its 15-member-per-regime, 10–200-day diagnostics adapt the forms of Bire et al. Figures 3, 5, 6, 7, and 9 to the 1-degree tutorial: multi-lead snapshots, spatial RMSE, ACC with training-global anomaly reference, and low/control/high wind-regime skill. This is explicitly a diagnostic-form comparison, not a numerical reproduction of the paper figures.
2026-07-21	Model A overfit submitted	After 48 passing project tests (one intentional dependency-path skip), style/diff checks, a 62-by-62 CPU forward/backward pass, and a production-Zarr sample read, submitted V100 job 284068. It ran the declared 96 regime-balanced, training-only pairs for the residual-learning/reload acceptance gate.
2026-07-21	Model A overfit passed	Job 284068 trained for 5m49s but failed only at checkpoint reload because NeuralOperator 2.0 adds a non-parameter <code>_metadata</code> entry that PyTorch rejects as a state-dict key. Filtered that entry from portable checkpoints, added a reload unit test, set <code>CUBLAS_WORKSPACE_CONFIG</code> for CUDA determinism, and reran the unchanged gate as job 284151. Job 284151 completed in 5m58s: the masked relative- L_2 loss fell from 0.52061 to 0.01589 (best), and save/reload was bitwise exact. Checkpoint SHA-256: 1a8ae303b797...43c21aeaad5b2.

Date	Type	Change
2026-07-21	Model A development submitted	After 52 passing project tests (one intentional dependency-path skip), submitted V100 job 284152. It trains only the 7,530 chronological train pairs and scores the 780 held validation pairs for 12 epochs, against persistence, to select a single final Model A training duration.
2026-07-21	Model A schedule frozen	Development job 284152 completed in 17m24s. Its best held validation relative- L_2 loss was 0.018119 versus persistence 0.110743 at epoch 10; final epoch loss was 0.019212, and reload was bitwise exact. Froze one final schedule (10 epochs, batch 8, learning rate 0.001, AdamW $\beta = (0.9, 0.95)$, 10^{-5} weight decay, seed 20260721) and submitted V100 job 284153.
2026-07-21	Model A model frozen	Final job 284153 completed in 14m38s. It preserved bitwise checkpoint reload and achieved held validation/inference relative- L_2 losses 0.018119/0.018400 versus persistence 0.110743/0.110719. Submitted job 284155 to save Model A one-step, rollout, snapshot, and direct A0-versus-A figures plus numerical outputs in the project-facing output tree.
2026-07-21	Model A evaluation complete	Job 284155 completed in 1m38s and saved metrics/arrays plus four figures under <code>outputs/af_fno/A/final_v1/</code> . Held 10-day Model A/persistence RMSE ratios are 0.169 (U), 0.467 (V), 2.052 (temperature), and 2.277 (SSH), compared with A0's 0.395, 0.912, 4.771, and 13.612. At 100 days Model A still has temperature/SSH ratios 17.79/24.48; therefore A is a forward-state improvement over A0 but not a stable long-rollout solution.
2026-07-22	Model B gate submitted	Implemented the controlled Model B forward-loss pipeline without changing Model A's architecture, data, residual target, optimizer, or seed. The loss contract adds 20/30-day autoregressive error, 12-bin radial anomaly spectra at 10/20/30 days, and the first four wet cells from the western wall at all three leads. Added incremental development profiles, frozen-protocol A0/A/B/persistence evaluation, loss-contract hashing, checkpoint reload gates, tests, and Slurm wrappers. All 62 available tests passed (one intentional skip), and a production-grid three-step CPU forward/backward check had finite gradients. Submitted the 96-sample full-loss V100 acceptance run as job 284302.
2026-07-22	Model B overfit passed	Job 284302 completed in 13m12s. On 96 balanced training-only rollouts it reduced complete loss from 1.52456 to 0.04213 and one-step state loss from 0.52075 to 0.01704; rollout, spectral, and western-boundary components all fell, learning rate 0.001 passed without fallback, and checkpoint reload was bitwise exact. Checkpoint SHA-256: 512707d82bce...17871c159e28. Dependency-gated rollout-only, rollout-plus-spectral, and full-loss development jobs 284304–284306 started after acceptance.

Date	Type	Change
2026-07-22	Model B schedule frozen	Development jobs 284304–284306 completed in 43–44 minutes with bitwise reload. At each profile’s held minimum, common full-objective values were 0.044899 (rollout only), 0.044081 (rollout plus spectra), and 0.043064 (full); the full profile also had the lowest state (0.017925), rollout (0.033599), spectral (0.009012), and boundary (0.031558) terms. Its minimum was epoch 12 versus persistence 0.306148. Froze epochs 12, batch 8, learning rate 0.001, unchanged optimizer/seed/loss weights, and submitted final job 284342. Full-development checkpoint SHA-256: <code>e185477b126c...af23a7a8119d</code> .
2026-07-23	Model B closed	Final job 284342 completed in 36m19s on one V100, preserved bitwise reload, and saved the frozen checkpoint. Evaluation job 284346 completed the common 10–100-day package. B reduces A’s 100-day U/V/temperature/SSH errors to 0.521/0.409/0.450/0.405 of A, but its own persistence ratios remain 0.449/0.997/8.009/9.925. B is retained unchanged as a successful rollout-loss ablation, not accepted as the forward parent of the response model.
2026-07-23	Complete jobs cancelled	Complete one-year A/B characterization jobs 284816 and 284817 were pending with zero runtime and were cancelled. Their frozen checkpoints and existing evidence remain; uniform final packages are deferred until an accepted C configuration freezes so they are generated once for the final comparison.
2026-07-23	Model ladder revised	Inserted forward-selected Model C between B and the response-supervised model, now named D. The Duruisseaux practical FNO workflow and Bire evidence pattern remain the references. C uses a bounded dense-FNO search and must pass the complete forward gate before response or adjoint work opens.
2026-07-23	Model C started	Added the bounded C architecture/loss contract and training-only diagnostics. Equal U/V/temperature/SSH state and boundary terms, per-channel ten-day increment RMS, and Hann-tapered increment spectra directly address B’s aggregate-loss and spectral weaknesses. Eight Model C tests, style checks, shell validation, and a 62×62 three-step forward/backward pass succeeded; width-32/64 parameter counts are 630,262/2,443,542. CPU job 284849 exposed an incorrect assertion that checked future pair-start codes rather than target snapshot codes and stopped before creating output. Added a regression test and submitted corrected job 284850 with unchanged train-only scope.

Date	Type	Change
2026-07-23	Model C C0 complete	Corrected job 284850 completed in 42s and verified the immutable dataset hash and train-only read contract. Its 96-sample audit found a 147-fold increment-RMS range and only 278–398 effective increment samples. Because extra meridional modes retained more velocity-increment energy than extra zonal modes, added (24, 16) before validation and reran job 284857 in 33s. At (16, 16), U/V/temperature/SSH increment fractions are 8.8/24.0/67.8/85.6%; at (24, 16), 12.5/32.9/70.2/89.0%.
2026-07-23	Model C C1a submitted	Added a training-only loss-gradient calibration: 20-epoch core warmup on 96 balanced rollouts, followed by raw-term and parameter-gradient audits. It proposes, but does not freeze, auxiliary weights and writes no candidate checkpoint. Nine Model C tests, style, shell, and production-grid checks passed; submitted V100 job 284858.
2026-07-23	CPU C1a replica corrected	CUDA is not scientifically required for the 96-rollout calibration. Job 284860 failed safely after 19s, before output, because its initial audit squared very large gradients in float32. Moving the global norm to float64 prevented overflow, but live job 284864 then revealed that a direct cast discarded imaginary spectral-weight gradients; it was cancelled before output. The final helper preserves real gradients in float64 and complex gradients in complex128 and is covered by a large real/complex regression test.
2026-07-23	Model C C1a complete	Complex-safe CPU job 285116 completed in 4m39s using 1.53 GB maximum RSS. After warmup, state/increment/rollout/spectral/boundary gradient norms were 1.965/830.5/6.484/40,763/7.411. Rejected the clipped 0.01 increment/spectral proposal and froze rounded unclipped coefficients 0.001/0.15/10 ⁻⁵ /0.065, whose weighted gradient ratios to state are 0.423/0.495/0.207/0.245. Report SHA-256: 3406c9104dd6...592e469. Cancelled redundant pending GPU job 284858 with zero runtime.
2026-07-23	Model C C1b submitted	Added a 96-rollout, training-only memorization gate with fixed initialization/batch order across the bounded learning-rate fallback. Acceptance is groupwise rather than aggregate: 90% total reduction, 50% spectral reduction, maximum state group 0.08, maximum increment group 1.0, rollout 0.10, boundary 0.15, and bitwise-exact three-step reload. Eleven Model C tests and all style/diff/shell checks passed; submitted eight-core short-queue job 285192.

Date	Type	Change
2026-07-23	Model C C1b rejected	Job 285192 completed both 160-epoch attempts in 58m30s and failed only the increment criterion. At learning rate 0.0005, total/spectral reductions were 97.7/99.1%, maximum state group was 0.0304, rollout/boundary were 0.0312/0.0397, and U/V increments were 0.254/0.442 times exact-subset persistence; temperature/SSH remained 1.285/2.032. The absolute-1 threshold is replaced by the mathematically correct per-group ratio to persistence on the same sampled records, without relaxing the requirement.
2026-07-23	Model C C1c submitted	The lower learning rate reached its best total at epoch 160, so undertraining is tested before changing objective or capacity. Added version-2 diagnostics that preserve full learning curves and a reload-tested best checkpoint on rejection. Job 285265 fixes data, architecture, loss, seed, batch order, and learning rate 0.0005, changing only duration from 160 to 320 epochs.
2026-07-23	Pressure evaluation completed	Added a fixed MITGCM-consistent PHIHYPD postprocessor without changing any model input, output, weight, or split. It integrates the configured linear-EOS density anomaly and adds the linear-free-surface $g\eta$ term. Against the archived S0 PH dump at iteration 2,851,200, all-depth wet-cell maximum error is 3.89×10^{-4} and RMSE is $9.75 \times 10^{-6} \text{ m}^2 \text{ s}^{-2}$. Evaluation contract v2 adds surface/mid/bottom pressure curves, maps, snapshots, spectra, boundary metrics, and gate comparisons. Existing v1 packages remain immutable and pre-pressure; frozen inference will be replayed once for the uniform v2 comparison after an accepted C configuration freezes.
2026-07-26	Model C C1c rejected	Job 285265 completed all 320 epochs and every total, spectral, state, rollout, boundary, and exact-reload criterion, but no evaluated epoch passed every increment group. Epoch 315 gave U/V/temperature/SSH persistence ratios 0.206/0.353/0.979/1.219. This is a scientific SSH rejection, although the scheduler recorded exit 1 under the older runner.
2026-07-26	Width-64 capacity rejected	GPU job 285325 completed 320 epochs with exact reload. Its selected epoch ratios were 0.145/0.292/1.083/1.385, and the best balanced epoch still had SSH at 1.302. Four times the parameter count did not solve the slow-field failure, so a further width increase is unsupported.
2026-07-26	CPU fallback measured	Scaling job 287556 completed the same short benchmark at 4, 8, and 16 CPU cores in 436, 272, and 253 seconds. Sixteen cores is the measured fallback if GPU access would materially delay useful work; backend replication remains secondary.

Date	Type	Change
2026-07-26	Late-objective audit complete	GPU array 287581 completed in 30/33 seconds and independently rechecked all 64 evaluated epochs in each width history, the exact 96 training records, reload provenance, and every memorization criterion. At widths 32/64, weighted increment loss was 5.7/7.2% of state but its gradient norm was already 0.422/0.406 times state; SSH dominated that gradient and was strongly aligned with state. Exact gradient reconstruction errors were approximately 10^{-7} .
2026-07-26	Memorization schedule accepted	Bounded GPU array 287583 isolated two changes. Task 1's versioned 0.0025 increment-weight loss was scientifically rejected, with best balanced SSH ratio 1.328. Task 0 retained loss v1 and applied learning rate 0.0005 through epoch 240 then 0.0001; epoch 315 passed every gate at U/V/temperature/SSH ratios 0.211/0.360/0.794/0.990, 98.24/99.82% total/spectral reductions, 0.0248 maximum state error, 0.0231 rollout, 0.0301 boundary, and exact reload. Only epoch 315 passed every group, so this authorizes bounded validation but does not validate or freeze C.
2026-07-26	Validation contract frozen	Before reading any validation state metric, froze <code>config/model_c_validation_search_v1.json</code> with SHA-256 <code>9e1d44299ae6...c4c6f6</code> . Ten mode/width candidates undergo four from-scratch rounds with 25/50/75/100% chronology and 1,920/3,840/5,760/7,680 optimizer steps. Ranking first requires all physical U/V/temperature/SSH ten-day ratios below persistence, then minimizes worst-group ratio and a 30/90/180-day full-domain/boundary/spectral score. Conditional padding/layer triggers and final seeds 20260723/20260724/20260725 are fixed; inference and later data remain sealed.
2026-07-26	Validation stages 1–2 complete	GPU array 287604 completed all ten 25%-chronology candidates in 5m58s–8m45s and advanced <code>m16_24_w64</code> , <code>m24_16_w64</code> , <code>m24_24_w64</code> , <code>m12_12_w64</code> , and <code>m12_12_w32</code> ; manifest SHA-256 <code>40b54530d7d9...12d2f05</code> . Job 288466 completed the five fresh 50% runs in 10m21s–15m15s and advanced <code>m16_24_w64</code> , <code>m24_16_w64</code> , and <code>m12_12_w64</code> ; their worst-group SSH ratios were 1.470, 1.483, and 1.599. Manifest SHA-256: <code>a2d639f8940c...d5e382</code> . Every scheduler task and exact-reload check passed; no candidate yet met the scientific gate.

Date	Type	Change
2026-07-26	Validation stages 3–4 complete	Job 289379 completed the three fresh 75% runs in 16m24s–21m51s and advanced <code>m16_24_w64/m24_16_w64</code> , whose SSH ratios were 1.153/1.192; manifest SHA-256 <code>9bbd4fc4ce3c...f30e17</code> . Job 289915 completed both fresh full-resource runs in 21m18s–28m37s. The (24, 16), width-64 design ranked first at U/V/temperature/SSH ratios 0.1473/0.3547/0.6597/1.00014 and physics score 22.84, ahead of (16, 24), width 64 at 0.1577/0.3723/0.6353/1.02477 and 28.22. Training worst-group ratios 0.915/0.941 and boundary-to-full ratios 0.667/0.639 left both conditional triggers false, so no depth or padding branch opened. Stage-4 manifest SHA-256: <code>453c05ec408f...2fea2</code> .
2026-07-26	Three-seed gate rejected	GPU array 290382 trained the selected (24, 16), width-64 architecture afresh for seeds 20260723/20260724/20260725; all tasks completed with exit 0 in 28m39s/22m23s/21m36s and all three checkpoints reloaded bitwise exactly. U/V/temperature beat persistence for every seed, but SSH ratios 1.00014/1.00383/1.03818 failed the strict every-seed rule. The immutable decision therefore records <code>scientifically_rejected_three_seed_gate</code> , <code>configuration_frozen=false</code> , and <code>inference_authorized=false</code> ; SHA-256 <code>e19d502442fc...72155</code> . Inference, I1/I2, response, and adjoint data remain sealed.
2026-07-27	Freeze data-adequacy gate	Before computing per-regime checkpoint metrics, froze <code>config/model_c_data_adequacy_v1.json</code> , SHA-256 <code>e185301947b4...53c84a9</code> . It requires aggregate training fit for every seed, a strictly improving four-stage chronology curve, a positive median SSH validation–training gap in at least two regimes, and no more than 20 effective state samples for both temperature and SSH. It reads only split codes 1/2.
2026-07-27	Authorize trajectory v2	GPU job 290415 completed in 3m16s. All four expansion checks passed; state-RMS effective totals are 7.59/7.69 for temperature/SSH, and aggregate SSH block-bootstrap intervals include persistence. S1 is the localized weakness: training ratios 1.32–1.36 and validation ratios 1.50–1.55, versus sub-persistence S0/S2. Report SHA-256 <code>be672face8c4...4e307ee</code> authorizes a two-times data expansion but also requires non-data successor changes.
2026-07-27	Freeze expansion design	<code>config/trajectories_v2_expansion.json</code> , SHA-256 <code>7ee2ae10330a...fe045ed</code> , declares exact ten-year continuations of S0–S2 in a new <code>mitgcm_v2/</code> root. The successor store has 7,200 records per regime and 15,060 training pairs; opened v1 validation is excluded, original inference is preserved, and fresh validation/inference blocks retain 90-day buffers. Effective coverage must be measured after conversion.

Date	Type	Change
2026-07-27	Complete v2 simulations	Array 290439 completed all three exact ten-year continuations with exit 0 in 15m10s–15m31s. Each task wrote 3,600 daily state pairs, a final pickup, a gap-free 72-iteration chronology, and STOP NORMAL END. No v1 artifact was overwritten.
2026-07-27	Validate trajectory v2	Jobs 290443/290444 built and independently checked the 11 GB $3 \times 7200 \times 46 \times 62 \times 62$ store. The v1 prefix and sampled raw extension records are bitwise exact; finite, land-zero, split, chronology, and train-only-normalizer checks pass. Dataset metadata/quality SHA-256 values are <code>ae7be47c4569...78b8cc/a6f7e5234ace...cf752</code> .
2026-07-27	Pass effective-coverage target	Contract SHA-256 <code>e539ca1bb61a...9197f36</code> was frozen after quality control and before metrics. Job 290446 found temperature/SSH state multipliers 2.142/2.039 and increment multipliers 2.106/2.253, passing the prospective two-times slow-field target. Report SHA-256: <code>7758cfb64e73...18e5</code> .
2026-07-27	Start C successor phase 1	Before v2 training metrics, froze contract SHA-256 <code>3e0a300e73ec...159285</code> . Loss, modes, depth, padding, optimizer exposure, and training records remain fixed. GPU array 290448 compares the exact width-64 data-only control with the isolated width-64 4C Channel-MLP ablation; the width-128/256-lift/256-projection candidate is phase 2. Fresh validation remains sealed.
2026-07-27	Complete C successor phase 1	Array tasks 290448_0/290448_1 completed with exit zero in 41m53s/57m49s. Both have exact three-step reload and finite, amplitude-bounded 180-day rollouts. Each fails only S1 SSH: the data-only control is 1.103435 times persistence and isolated 4C channel mixing is 1.003573 over all 5,020 S1 training pairs. The 4C change improves aggregate U/V/temperature/SSH ratios by 23.6/22.2/9.4/7.9%, so it is beneficial but not sufficient under the exact gate. Reports/checkpoints are hash-addressed in <code>outputs/af_fno/C/successor_training_v1/phase_1/</code> .
2026-07-27	Start C successor phase 2	Because both phase-1 reports completed operationally, submitted the already-contracted width-128, lift-256, projection-256, 4C candidate as GPU job 290597. It changes no data, loss, modes, depth, padding, optimizer exposure, seed, or training records and does not read fresh validation or any later archive.
2026-07-27	Pass C successor training gate	Job 290597 completed with exit zero in 1h20m22s. Aggregate U/V/temperature/SSH ratios are 0.049040/0.121279/0.384816/0.522413; every group passes separately in S0/S1/S2, with S1 SSH at 0.762683. Exact reload, finite 180-day rollout, and every 90/180-day amplitude check pass. Report/checkpoint SHA-256 values are <code>5946515ea81a...68433c/2d10e48adc5e...360060</code> .

Date	Type	Change
2026-07-27	Freeze successor validation v2	Before any fresh validation state metric, froze seeds 20260723/24/25, all complete 10–90-day starts, Bire-primary surface speed/SST/derived pressure, persistence/climatology/A0 baselines, paired block-bootstrap confidence, and SSH/streamfunction stability. A split-only audit found exactly 180 complete starts per regime, indices 6300–6479; immutable v1 expected 181 and is superseded by v2 SHA-256 56bb6ec59799...0cf27b without changing science.
2026-07-27	Start successor seed replication	GPU array 290673 trains only the missing seeds 20260724/20260725 with the accepted width-128 architecture and identical training-only checkpoint-selection rule. Seed 20260723 reuses job-290597’s hash-pinned checkpoint. Split 2 and every later archive remain unread.
2026-07-27	Pass three-seed training gate	Array tasks 290673_0/290673_1 completed with exit zero in 1h08m03s/1h09m54s. Seeds 20260723/20260724/20260725 all beat persistence for every U/V/temperature/SSH group separately in S0/S1/S2; their worst ratios are 0.762683/0.732270/0.839507, each for S1 SSH. Every exact-reload, finite-rollout, and 90/180-day amplitude criterion passes. This authorizes one execution of frozen fresh-validation v2, not inference.
2026-07-27	Start frozen fresh validation	Submitted GPU job 290738 to apply the unmodified v2 contract once to all three hash-pinned checkpoints and all 180 complete 10–90-day starts per regime. Split 2 is now authorized for this job only; inference, intermediate winds, response, and adjoint data remain sealed.
2026-07-27	Reject successor fresh validation	Job 290738 completed with exit zero in 13m20s, but no seed passed the full point gate and the paired bootstrap failed. Three-seed mean surface-speed RMSE-AUC is 0.245 times persistence and passes every baseline. SST/surface-pressure ratios are 2.313/2.198: both are good at day 10, cross persistence around days 20–30, and reach about 4.5–5 by day 90. Every seed is finite, land-zero, and passes SSH/streamfunction amplitude and wind-slope checks. The normalized-maximum limit 20 also fails, but validation truth reaches 21.092, so that subgate is non-discriminating; primary failures independently reject C. Report/arrays SHA-256: 9853ea7aff21...663fb/fd84199b585e...dccc.
2026-07-27	Freeze rollout diagnosis v1	Before reading balanced long-rollout training metrics, froze config/model_c_rollout_diagnosis_v1.json, SHA-256 d0ecee0db2b2...ff7ba. It evaluates the three checkpoints on 90 evenly spaced complete starts from each of the two training blocks, 540 records across regimes, at 10–90 days. It changes no architecture/loss and reads no validation states or split-3/later archive.

Date	Type	Change
2026-07-27	Start rollout diagnosis	Submitted GPU job 291102 to execute the frozen 540-record split-1 audit. Its only purpose is to classify the objective/checkpoint-gate mismatch before any bounded successor revision; it cannot authorize inference.
2026-07-27	Complete rollout diagnosis	Job 291102 completed with exit zero in 3m58s. Every seed reproduces good day-10 slow-field skill followed by failed 10–90-day RMSE-AUC and day-90 skill on balanced training chronology. Mean day-10/AUC/day-90 persistence ratios are 0.809/2.021/4.221 for SST, 0.552/1.867/3.873 for surface pressure, and 0.532/1.844/3.835 for SSH. Apply the frozen objective/checkpoint-gate-mismatch classification; report/arrays SHA-256 values are 7efdb0726125...171374/c3c403ab81b2...5f255.
2026-07-27	Save validation figures	Generated five project-facing PNGs from the immutable job-290738 member arrays and saved them beside the lightweight validation summary. Manifest content SHA-256 8b24b6af0686...a0845f binds the figures to report/array SHA-256 values 9853ea7aff21...663fb/fd84199b585e...dccc; no sealed archive was opened.
2026-07-27	Save diagnosis figures	Generated three hash-bound project-facing figures, including a direct training-versus-validation slow-field comparison, plus a lightweight summary under <code>outputs/af_fno/C/rollout_diagnosis_v1/</code> . Manifest content SHA-256 is 664e66a0866d...393671; no sealed archive was opened.
2026-07-27	Classify two-input/two-output option	Confirmed that current C is one-state-in/one-residual-out in time; its 51/46 figures are physical channel counts. Samudra used two consecutive states to predict two future states, whereas Bire named that method as future work. Preserve it as a bounded temporal ablation compared against the smaller long-rollout-objective correction rather than adopted without evidence.
2026-07-27	Freeze Bire-style figures	Before reading any 100–200-day validation metric, froze contract SHA-256 791c84f12c2d...eb696b. It fixes one seed-20260723 checkpoint, 15 randomly drawn complete S2 validation starts, the first draw for the spatial example, native-1° streamfunction truth/prediction/difference panels, and $\Delta t = 10$ RMSE curves through day 200. Persistence repeats the initial state; climatology uses all split-1 pointwise training means. The characterization cannot tune or authorize inference.
2026-07-27	Start Bire-style figures	Submitted GPU job 291134 under the unchanged contract. It reads only split-1 training states for climatology and the already opened S2 fresh-validation states for the 15 fixed rollouts; inference and all later archives remain sealed.

Date	Type	Change
2026-07-27	Complete Bire-style figures	Job 291134 completed with exit zero in 51 seconds. The fixed member's day-10/20/30/40 streamfunction RMSE is 0.0316/0.0934/0.1580/0.1765 Sv against about 13.34 Sv truth RMS. All 15 rollouts are finite, but day-200 Model-C/persistence RMSE is 0.07344/0.00480 m s^{-1} for speed, 5.981/0.0505 $^{\circ}\text{C}$ for SST, and 3.747/0.0253 $\text{m}^2 \text{s}^{-2}$ for surface pressure. The requested SST/pressure axes clip the model mean from days 100/160. Preserve report/arrays SHA-256 b704b921444d...e300ac5/a73ac9bf2cc9...20b1 and output manifest-content SHA-256 258be6fabe13...827f; no inference data were opened.
2026-07-27	Freeze per-lead streamfunction maps	Before evaluating the new maps, froze contract SHA-256 ccb965e58bdc...ae1934c. It reuses the fixed S2 start 6335 and checkpoint, generates exactly eight separate day-20–90 truth/prediction/difference figures, fixes one shared state scale, and requires a labeled lead-specific difference scale with numerical RMSE/max error. The original output remains immutable; inference remains sealed.
2026-07-27	Complete per-lead streamfunction maps	GPU job 291135 completed with exit zero in 35 seconds and all eight requested PNGs passed hash checks. RMSE/max error grows from 0.0934/0.7240 Sv at day 20 to 0.2951/1.1713 Sv at day 90; relative RMSE remains 0.700–2.211%. Days 20–40 reproduce the original maps to FP32 roundoff. Preserve report/arrays SHA-256 71c6e1dc1f9a...89c206/b4126494d9c0...3bcd48 and manifest-content SHA-256 3b74d759d551...450ac7; inference remains sealed.
2026-07-27	Save Figure-4 baseline companion	Generated a full-range/baseline-zoom SST and surface- P/ρ figure and a complete CSV from only the immutable job-291134 arrays. SST loses both baselines at day 20; pressure loses persistence at day 30 and climatology at day 70. Contract/manifest-content SHA-256 values are 2a2324090b83...b2a4a/01433936e406...99063; no model or dataset was read.
2026-07-27	Freeze and start exact checkpoint replay	Frozen contract SHA-256 494ce9de6688...06682 replays the original seed-20260723 optimization without changing architecture, loss, optimizer, batch order, or exposure. It saves all six late checkpoints, requires bitwise equality at the original step 14,880, and scores fixed split-1 SST/ P/ρ curves against both baselines before permitting either checkpoint-only selection or one bounded rollout-objective test. Preflight, lint, shell validation, and 20 related tests pass. Submitted only this gate as GPU job 291164; no objective variant is queued.

Date	Type	Change
2026-07-28	Complete exact checkpoint replay	Job 291164 completed normally in 1h19m14s. The original selected state is bitwise exact and the six training records have zero numerical difference. All checkpoints are finite and land-zero, but none passes the long-horizon gate. Step 14,400 is diagnostic best with worst primary AUC ratio 2.366 and worst slow-field lead ratio 5.513; the result requires objective correction.
2026-07-28	Start bounded pushforward correction	Frozen contract SHA-256 406ee5248fb4...bccc2 makes one change from step 14,400: alternating detached day-60/day-90 SST and surface- P/ρ terminal supervision with coefficient 0.0025. Width, modes, layers, padding, normalization, loss-v1 core, and split-1 gate are unchanged. Sixteen related tests and preflight pass; V100 job 291612 is running and no replica or sealed-state job is queued.
2026-07-28	Reject first push-forward gate	Job 291612 completed normally in 20m37s and selected reload-exact step 1,920. Worst primary AUC improves from 2.366 to 1.692, worst slow-field lead ratio from 5.513 to 3.994, and ten-day worst regime/group ratio from 0.852 to 0.610, but neither SST nor surface pressure beats both baselines through day 90. Publication manifest-content SHA-256 is 2b7cc78262e6...fa272; no replication or validation.
2026-07-28	Start replay-verified duration extension	All pushforward training terms are still descending and the low learning rate covered only the final 480 steps. Frozen contract SHA-256 4867fa77d68e...78c79 replays the complete path and requires bitwise step-1,920 equality before adding 3,840 steps at 2×10^{-5} . Nine related tests, lint, shell validation, and preflight pass; V100 job 291616 is running and all sealed reads remain blocked.
2026-07-28	Reject pushforward duration	Job 291616 completed normally in 51m31s and reproduced pushforward-v1 step 1,920 bitwise. All eight checkpoints fail. Diagnostic-best step 5,760 improves worst primary AUC/slow-lead ratios to 1.597/3.609, but day-90 SST is 2.402/3.609 and surface- P/ρ 2.922/1.727 times persistence/climatology. Published manifest-content SHA-256 is e97e78d5729c...e0f51; replication and validation remain blocked.
2026-07-28	Correct and start three-call truncated unroll	Initial job 291769 stopped before training or metrics because the source validator expected <code>fine_tune_step</code> , while the duration checkpoint records <code>total_fine_tune_step</code> . Version-2 contract SHA-256 0c5aabadbe51...92e3c validates the actual schema during preflight and changes no scientific setting. Four focused tests, lint, shell validation, source hashes, and preflight pass; V100 job 291778 is running on node112.

Date	Type	Change
2026-07-28	Reject three-call truncated unroll	Job 291778 completed normally in 20m01s and diagnostic-best step 1,920 reloaded a nine-step rollout bitwise. Every check-point fails; worst primary AUC/slow-lead ratios 1.933/3.701 are worse than duration step 5,760 at 1.597/3.609. Day-90 SST is 2.463/3.701 and surface- P/ρ 2.844/1.681 times persistence/climatology. Published figure/CSV/summary manifest-content SHA-256 is <code>cccd8e757495...ae4c03</code> . Stop automatic variants and keep validation sealed.
2026-07-28	Freeze slow-field bias/projection audit	Contract SHA-256 <code>fd7d39f5c6a4...05428</code> fixes the seed-20260723 width-128 checkpoint, all 5,020 split-1 teacher-forced pairs per regime, the exact 540 job-291102 starts, five post-hoc correction variants, five SST/SSH increment EOFs, damped persistence, and 720-day training decorrelation. Static bias is declared dominant only if fixed 9ϵ explains at least half the pooled SST and SSH day-90 error energy; a 20% worst-slow-AUC reduction is the alternate material-correction threshold. No held state or model weight may change.
2026-07-28	Complete stream-function side study	V1 verified the project centered-U operator exactly but its 10.9% U/V comparison placed U and V on different C-grid faces. Preserved that result as a warning, froze corrected v2 SHA-256 <code>88ff2f20fe1a...dea170</code> with unchanged thresholds, and compared raw-face prefix integrals at common corner indices. S0–S2 all pass: daily p90 relative RMSE is at most 3.25×10^{-5} , annual RMSE is $4.74\text{--}6.00 \times 10^{-5}$ Sv, correlations are effectively one, and boundary-normal transport is zero. Retain the free-surface caveat.
2026-07-27	Align final forward gate with Bire	Before fresh v2 validation, separate the strict all-group training-capacity test from final forward selection. Bire’s surface speed, SST, and derived surface pressure use 10–90-day RMSE/ACC curve integrals against persistence/climatology; streamfunction carries circulation/stability and wind-response physics. SSH remains mandatory with bootstrap and climate-stability diagnostics but is not a standalone veto, because Bire neither predicted nor scored it separately and surface pressure already includes $g\eta$.
2026-07-28	Complete slow-field bias/projection audit	Job 292064 completed normally in 7m09s on the frozen 540 split-1 starts and opened no held archive. Fixed 9ϵ explains 33.43%/27.21% of pooled SST/SSH day-90 error energy; basin mean plus five increment EOFs explain 4.57%/1.73%. Bias plus constraints reduces worst slow-field AUC from 1.986 to 1.603 (19.25%), below the frozen 20% threshold, while conservation alone slightly worsens it. Apply the feedback-amplification branch. Report/arrays SHA-256: <code>e66b263c826c...0f90/4e7f73c2384e...8547</code> .

Date	Type	Change
2026-07-28	Freeze rollout-conditioned loss v3	Contract SHA-256 <code>37f1157851ac...5a08</code> starts at duration step 5,760 and keeps the width-128 one-state interface, loss-v1 core, data order, and 540-record gate. It applies exact SSH and regime/level temperature-mean projections after every call, adds a weight-0.01 per-cell batch-mean slow-increment penalty, and cycles a weight-0.0025 one-call forecast-age loss over days 10–90. Five focused tests, lint, shell validation, source hashes, exact forcing-regime identification, and sealed-data preflight pass; no replica or held-state job is authorized.
2026-07-28	Complete rollout-conditioned loss v3	GPU job 292293 completed normally in 27m46s without opening held states. Diagnostic step 960 reloads exactly and all ten-day groups beat persistence at U/V/temperature/SSH ratios 0.0382/0.1058/0.2746/0.3456. The strict long gate fails: worst primary AUC improves from 1.597 to 1.462, but worst slow-field lead is 3.634. SST and surface- P/ρ still reach 2.419 and 3.019 times persistence at day 90. Their day-10/day-30 values, surface speed, and streamfunction beat both persistence and climatology, so freeze the separate horizon-matched gate next; do not replicate or launch another unconstrained variant.